Sales Data Pipeline using Azure Databricks and Azure Data Factory

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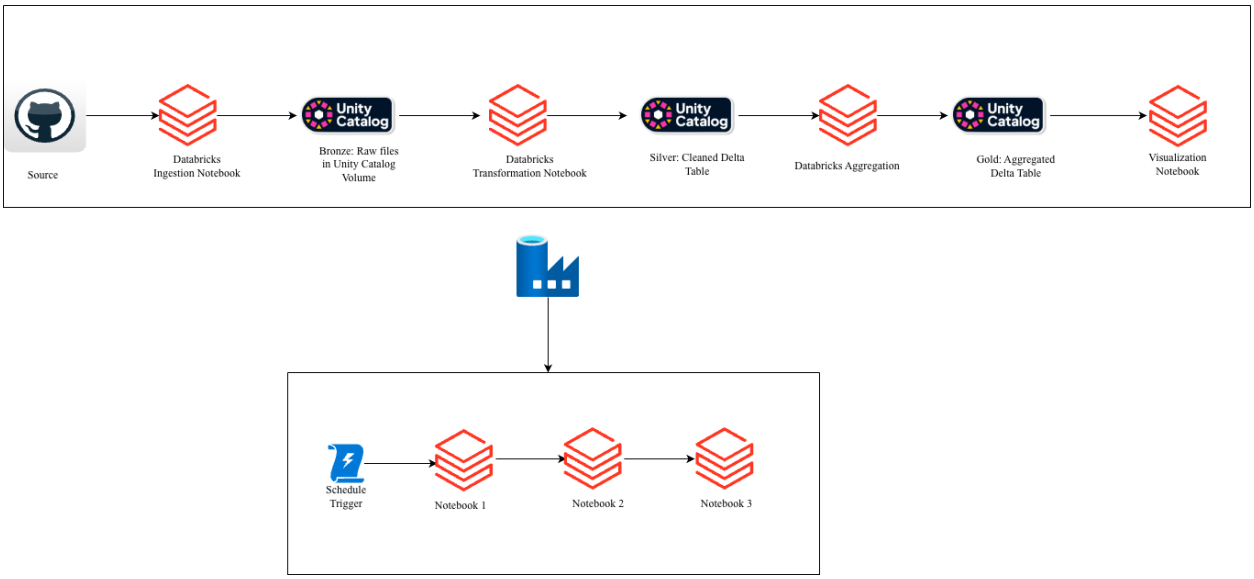
Objective:

The objective of this project is to build an end-to-end data pipeline using Azure Databricks and Azure Data Factory to ingest raw sales data from GitHub, process it through bronze, silver, and gold layers using a medallion architecture, and generate meaningful business insights through aggregation and visualization using Plotly.

Architecture:



Bronze Layer:

Create a catalog

Due to Unity Catalog metastore storage configuration limitations in the environment, separate catalogs could not be created. Therefore, the medallion architecture was implemented using separate schemas within a single catalog while maintaining proper layer separation.

So used the existing one

2 minutes ago (10s)

```
%sql
SHOW CATALOGS;
```

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [catalog: string]

Table

	catalog
1	rithikadb
2	samples
3	system

Create Schema

1 minute ago (1s)

```
%sql
CREATE SCHEMA IF NOT EXISTS rithikadb.bronze_schema;
```

OK

Create volume

Just now (2s)

```
%sql
CREATE VOLUME IF NOT EXISTS rithikadb.bronze_schema.raw_volume;
```

OK

Just now (<1s)

```
#Defined volume path
volume_path = "/Volumes/rithikadb/bronze_schema/raw_volume/"
```

4 days ago (<1s) 5

```
#Imported pandas
import pandas as pd

#loaded the url
urls = {
    "2019": "https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/2019.csv",
    "2020": "https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/2020.csv",
    "2021": "https://raw.githubusercontent.com/MicrosoftLearning/mslearn-databricks/main/data/2021.csv"
}
```

4 days ago (3s) 6

```
for year, url in urls.items():
    pdf = pd.read_csv(url)   #read using pandas
    df = spark.createDataFrame(pdf)  #create dataframe using spark

    df.write.mode("overwrite").csv(volume_path + f"{year}") #write it into volume using overwrite

print("Bronze ingestion complete")
```

(3) Spark Jobs

df: pyspark.sql.connect.dataframe.DataFrame = [SO49171: string, 1: long ... 7 more fields]

pdf: pandas.core.frame.DataFrame = [SO49171: object, 1: int64 ... 7 more fields]

```
Bronze ingestion complete
```

4 days ago (1s) 7

```
#Display all the files in the location
display(dbutils.fs.ls(volume_path))
```

(1) Spark Jobs

Table +

	path	name	size	modificationTime
1	dbfs:/Volumes/rithikadb/bronze_schema/raw_volume/2019/	2019/	0	1774440645000
2	dbfs:/Volumes/rithikadb/bronze_schema/raw_volume/2020/	2020/	0	1774440646000
3	dbfs:/Volumes/rithikadb/bronze_schema/raw_volume/2021/	2021/	0	1774440648000

3 rows | 0.82s runtime

4 days ago (3s) 8

```
df_2019 = spark.read.csv(volume_path + "2019", header=True, inferSchema=True)
display(df_2019)
```

(4) Spark Jobs

df_2019: pyspark.sql.connect.dataframe.DataFrame = [SO43704: string, 11: integer ... 7 more fields]

Table +

	SO43704	11	2019-07-01	Julio Ruiz	julio1@adventure-works.com	Mountain-100 Black, 48	16	3374.99
1	SO43705	1	2019-07-01	Curtis Lu	curtis9@adventure-works.com	Mountain-100 Silver, 38	1	339
2	SO43700	1	2019-07-01	Ruben Prasad	ruben10@adventure-works.com	Road-650 Black, 62	1	699.0
3	SO43703	1	2019-07-01	Albert Alvarez	albert7@adventure-works.com	Road-150 Red, 62	1	357
4	SO43697	1	2019-07-01	Cole Watson	cole1@adventure-works.com	Road-150 Red, 62	1	357
5	SO43699	1	2019-07-01	Sydney Wright	sydney61@adventure-works.com	Mountain-100 Silver, 44	1	339
6	SO43702	1	2019-07-01	Colin Anand	colin45@adventure-works.com	Road-150 Red, 44	1	357
7	SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventure-works.com	Mountain-100 Silver, 44	1	339
8	SO43707	1	2019-07-02	Emma Brown	emma3@adventure-works.com	Road-150 Red, 48	1	357
9	SO43711	1	2019-07-02	Courtney Edwards	courtney1@adventure-works.com	Road-150 Red, 56	1	357
10	SO43706	1	2019-07-02	Edward Brown	edward26@adventure-works.com	Road-150 Red, 48	1	357
11	SO43708	1	2019-07-02	Brad Deng	brad2@adventure-works.com	Road-650 Red, 52	1	699.0
12	SO43709	1	2019-07-02	Martha Xu	martha12@adventure-works.com	Road-150 Red, 52	1	357
13	SO43710	1	2019-07-02	Katrina Raji	katrina20@adventure-works.com	Road-150 Red, 56	1	357
14	SO43712	1	2019-07-02	Abigail Henderson	abigail73@adventure-works.com	Road-150 Red, 44	1	357
15	SO43720	1	2019-07-03	Melanie Sanchez	melanie47@adventure-works.com	Road-150 Red, 44	1	357

1,196 rows | 3.48s runtime Refreshed 4 days ago

4 days ago (1s) 9

```
df_2020 = spark.read.csv(volume_path + "2020", header=True, inferSchema=True)
display(df_2020)
```

(4) Spark Jobs

df_2020: pyspark.sql.connect.dataframe.DataFrame = [SO48182: string, 11: integer ... 7 more fields]

Table +

	SO48182	11	2020-10-16	Michelle Gray	michelle7@adventure-works.com	Mountain-200 Black, 42	16	20
1	SO48179	1	2020-10-16	Logan Mitchell	logan37@adventure-works.com	Road-250 Red, 48	1	
2	SO48177	1	2020-10-16	Gary Dominguez	gary23@adventure-works.com	Road-550-W Yellow, 42	1	
3	SO48180	1	2020-10-16	Kelli Hu	kelli20@adventure-works.com	Road-250 Black, 58	1	
4	SO48183	1	2020-10-16	Troy Sara	troy11@adventure-works.com	Road-250 Black, 44	1	
5	SO48184	1	2020-10-16	Marc Blanco	marc19@adventure-works.com	Road-550-W Yellow, 38	1	
6	SO48178	1	2020-10-16	Diana Ortega	diana21@adventure-works.com	Mountain-200 Silver, 38	1	
7	SO48181	1	2020-10-16	Blake Mitchell	blake38@adventure-works.com	Road-250 Red, 52	1	
8	SO48185	1	2020-10-17	Nathan Jenkins	nathan2@adventure-works.com	Mountain-200 Black, 42	1	
9	SO48188	1	2020-10-17	Kara Lal	kara9@adventure-works.com	Road-250 Black, 52	1	
10	SO48187	1	2020-10-17	Marshall Sutton	marshall25@adventure-works.com	Road-650 Black, 60	1	
11	SO48186	1	2020-10-17	Deanna Garcia	deanna18@adventure-works.com	Road-250 Black, 58	1	
12	SO48189	1	2020-10-17	Cameron Bryant	cameron13@adventure-works.com	Road-650 Red, 62	1	
13	SO48190	1	2020-10-17	Kathryn She	kathryn0@adventure-works.com	Road-650 Red, 60	1	
14	SO48196	1	2020-10-18	Devin Green	devin28@adventure-works.com	Road-650 Red, 44	1	
15	SO48197	1	2020-10-18	Pedro Suri	pedro0@adventure-works.com	Road-650 Red, 60	1	

2,728 rows | 1.42s runtime Refreshed 4 days ago

4 days ago (2s) 10

```
df_2021 = spark.read.csv(volume_path + "2021", header=True, inferSchema=True)
display(df_2021)
```

(4) Spark Jobs

df_2021: pyspark.sql.connect.dataframe.DataFrame = [SO49172: string, 11: integer ... 7 more fields]

Table +

	SO49172	11	2021-01-01	Brian Howard	brian23@adventure-works.com	Road-250 Red, 44	16	2443.35
1	SO49173	1	2021-01-01	Linda Alvarez	linda19@adventure-works.com	Mountain-200 Silver, 38	1	2071.4
2	SO49174	1	2021-01-01	Gina Hernandez	gina4@adventure-works.com	Mountain-200 Silver, 42	1	2071.4
3	SO49178	1	2021-01-01	Beth Ruiz	beth4@adventure-works.com	Road-550-W Yellow, 44	1	1000.4
4	SO49179	1	2021-01-01	Evan Ward	evan13@adventure-works.com	Road-550-W Yellow, 38	1	1000.4
5	SO49175	1	2021-01-01	Margaret Guo	margaret24@adventure-works.com	Road-250 Red, 52	1	244
6	SO49180	1	2021-01-01	Mitchell Yuan	mitchell6@adventure-works.com	Road-650 Black, 58	1	78
7	SO49176	1	2021-01-01	Shawn Sharma	shawn11@adventure-works.com	Mountain-200 Silver, 46	1	2071.4
8	SO49177	1	2021-01-01	Barbara Chande	barbara44@adventure-works.com	Mountain-200 Silver, 42	1	2071.4
9	SO49186	1	2021-01-02	Cara Xu	cara8@adventure-works.com	Road-250 Red, 52	1	244
10	SO49187	1	2021-01-02	Lacey Liu	lacey16@adventure-works.com	Road-250 Black, 58	1	2181.5
11	SO49190	1	2021-01-02	Omar Zhu	omar13@adventure-works.com	Road-550-W Yellow, 40	1	1000.4
12	SO49185	1	2021-01-02	Cassandra Fernandez	cassandra17@adventure-works.com	Mountain-200 Black, 46	1	2049.0
13	SO49184	1	2021-01-02	Monica Martinez	monica17@adventure-works.com	Mountain-200 Black, 46	1	2049.0
14	SO49189	1	2021-01-02	Marie Gonzalez	marie20@adventure-works.com	Road-650 Black, 48	1	78
15	SO49182	1	2021-01-02	Alexandra Hall	alexandra89@adventure-works.com	Road-250 Red, 48	1	244

10,000+ rows | Truncated data | 1.91s runtime Refreshed 4 days ago

Silver Layer:

Create schema for silver layer

4 days ago (8s)

```
%sql
CREATE SCHEMA IF NOT EXISTS rithikadb.silver_schema;
```

OK

Define the volume path and read all the csv

4 days ago (10s) 2

```python
volume_path = "/Volumes/rithikadb/bronze_schema/raw_volume/"

df_2019 = spark.read.csv(volume_path + "2019", header=True, inferSchema=True)
df_2020 = spark.read.csv(volume_path + "2020", header=True, inferSchema=True)
df_2021 = spark.read.csv(volume_path + "2021", header=True, inferSchema=True)
```

(6) Spark Jobs

```
df_2019: pyspark.sql.connect.dataframe.DataFrame = [SO43704: string, 11: integer ... 7 more fields]
df_2020: pyspark.sql.connect.dataframe.DataFrame
    SO48182: string
    11: integer
    2020-10-16: date
    Michelle Gray: string
    michelle7@adventure-works.com: string
    Mountain-200 Black, 42: string
    16: integer
    2049.0982: double
    163.9279: double
df_2021: pyspark.sql.connect.dataframe.DataFrame = [SO49172: string, 11: integer ... 7 more fields]
```

Print the no.of rows and the column heading of all the df

4 days ago (1s) 3

```python
print("Total rows in 2019:", df_2019.count())
print("Total rows in 2020:", df_2020.count())
print("Total rows in 2021:", df_2021.count())
```

(6) Spark Jobs

```
Total rows in 2019: 1196
Total rows in 2020: 2728
Total rows in 2021: 28779
```

4 days ago (<1s) 4 Python

```python
print(df_2019.columns)
print(df_2020.columns)
print(df_2021.columns)
```

```
['SO43704', '11', '2019-07-01', 'Julio Ruiz', 'julio1@adventure-works.com', 'Mountain-100 Black, 48', '16', '3374.99', '269.9992']
['SO48182', '11', '2020-10-16', 'Michelle Gray', 'michelle7@adventure-works.com', 'Mountain-200 Black, 42', '16', '2049.0982', '163.9279']
['SO49172', '11', '2021-01-01', 'Brian Howard', 'brian23@adventure-works.com', 'Road-250 Red, 44', '16', '2443.35', '195.468']
```

So create columns for all the df

4 days ago (<1s) 5

```
columns = [
    "order_id",
    "quantity",
    "order_date",
    "customer_name",
    "email",
    "product",
    "units",
    "sales_amount",
    "profit"
]
```

4 days ago (<1s) 6

```
df_2019 = df_2019.toDF(*columns)
df_2020 = df_2020.toDF(*columns)
df_2021 = df_2021.toDF(*columns)
```

```
df_2019: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 7 more fields]
df_2020: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 7 more fields]
df_2021: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 7 more fields]
```

4 days ago (<1s) 7

```
print(df_2019.columns)
print(df_2020.columns)
print(df_2021.columns)
```

```
['order_id', 'quantity', 'order_date', 'customer_name', 'email', 'product', 'units', 'sales_amount', 'profit']
['order_id', 'quantity', 'order_date', 'customer_name', 'email', 'product', 'units', 'sales_amount', 'profit']
['order_id', 'quantity', 'order_date', 'customer_name', 'email', 'product', 'units', 'sales_amount', 'profit']
```

Union all table by name

4 days ago (<1s) 9

```
df_all = df_2019.unionByName(df_2020).unionByName(df_2021)
```

df_all: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 8 more fields]

+ Code + Text Genie Code

4 days ago (4s) 10

```
display(df_all)
print("Total rows:", df_all.count())
```

(5) Spark Jobs

Table +

	order_id	quantity	order_date	customer_name	email	product	units	sales_amount
1	SO43705	1	2019-07-01	Curtis Lu	curtis9@adventure-works.com	Mountain-100 Silver, 38	1	339
2	SO43700	1	2019-07-01	Ruben Prasad	ruben10@adventure-works.com	Road-650 Black, 62	1	699.0
3	SO43703	1	2019-07-01	Albert Alvarez	albert7@adventure-works.com	Road-150 Red, 62	1	357
4	SO43697	1	2019-07-01	Cole Watson	cole1@adventure-works.com	Road-150 Red, 62	1	357
5	SO43699	1	2019-07-01	Sydney Wright	sydney61@adventure-works.com	Mountain-100 Silver, 44	1	339
6	SO43702	1	2019-07-01	Colin Anand	colin45@adventure-works.com	Road-150 Red, 44	1	357
7	SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventure-works.com	Mountain-100 Silver, 44	1	339
8	SO43707	1	2019-07-02	Emma Brown	emma3@adventure-works.com	Road-150 Red, 48	1	357
9	SO43711	1	2019-07-02	Courtney Edwards	courtney1@adventure-works.com	Road-150 Red, 56	1	357
10	SO43706	1	2019-07-02	Edward Brown	edward26@adventure-works.com	Road-150 Red, 48	1	357
11	SO43708	1	2019-07-02	Brad Deng	brad2@adventure-works.com	Road-650 Red, 52	1	699.0
12	SO43709	1	2019-07-02	Martha Xu	martha12@adventure-works.com	Road-150 Red, 52	1	357
13	SO43710	1	2019-07-02	Katrina Raji	katrina20@adventure-works.com	Road-150 Red, 56	1	357
14	SO43712	1	2019-07-02	Abigail Henderson	abigail73@adventure-works.com	Road-150 Red, 44	1	357
15	SO43720	1	2019-07-03	Melanie Sanchez	melanie47@adventure-works.com	Road-150 Red, 44	1	357

10,000+ rows | Truncated data | 4.02s runtime Refreshed 4 days ago

```
Total rows: 32703
```

4 days ago (<1s) 11

```
# drop duplicates
df_all = df_all.dropDuplicates()
```

df_all: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 8 more fields]

4 days ago (<1s) 12

```
# fill null values
df_all = df_all.fillna({
    "sales_amount": 0,
    "profit": 0
})
```

df_all: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 8 more fields]

4 days ago (<1s) 13

```python
# added order_year and order_month
from pyspark.sql.functions import year, month

df_all = df_all.withColumn("order_year", year("order_date"))
df_all = df_all.withColumn("order_month", month("order_date"))
```

df_all: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 10 more fields]

4 days ago (4s)

```python
df_all.printSchema()
print("Total rows:", df_all.count())
display(df_all)
```

(5) Spark Jobs

```
root
 |-- order_id: string (nullable = true)
 |-- quantity: integer (nullable = true)
 |-- order_date: date (nullable = true)
 |-- customer_name: string (nullable = true)
 |-- email: string (nullable = true)
 |-- product: string (nullable = true)
 |-- units: integer (nullable = true)
 |-- sales_amount: double (nullable = false)
 |-- profit: double (nullable = false)
 |-- year: integer (nullable = false)
 |-- order_year: integer (nullable = true)
 |-- order_month: integer (nullable = true)

Total rows: 32703
```

Saved the delta table

4 days ago (11s)

```
df_all.write.format("delta") \
    .mode("overwrite") \
    .saveAsTable("rithikadb.silver_schema.cleaned_data")
```

(2) Spark Jobs

Gold Layer:

Create a schema

4 days ago (1s)

```
%sql
CREATE SCHEMA IF NOT EXISTS rithikadb.gold_schema;
```

OK

Load the delta table from silver

4 days ago (3s) 2

```
df = spark.read.table("rithikadb.silver_schema.cleaned_data")
display(df)
```

(2) Spark Jobs

df: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, quantity: integer ... 10 more fields]

Table +

	order_id	quantity	order_date	customer_name	email	product	units	sales_amount
1	SO43759	1	2019-07-08	Jon Chander	jon12@adventure-works.com	Road-150 Red, 62	1	3578.
2	SO44037	1	2019-07-29	Edwin Xu	edwin13@adventure-works.com	Road-650 Red, 48	1	699.09
3	SO43805	1	2019-07-14	Samuel Hernandez	samuel49@adventure-works.com	Road-650 Red, 48	1	699.09
4	SO43832	1	2019-07-16	Isaiah Cox	isaiah9@adventure-works.com	Road-150 Red, 62	1	3578.
5	SO43932	1	2019-07-19	Alan Xu	alan15@adventure-works.com	Road-150 Red, 56	1	3578.
6	SO43715	1	2019-07-03	Warren Jai	warren42@adventure-works.com	Road-150 Red, 56	1	3578.
7	SO43954	1	2019-07-21	Virginia Mehta	virginia16@adventure-works.com	Road-150 Red, 44	1	3578.
8	SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventure-works.com	Mountain-100 Silver, 44	1	3399.
9	SO43986	1	2019-07-24	Ian Morris	ian85@adventure-works.com	Road-150 Red, 62	1	3578.
10	SO43973	1	2019-07-23	Xavier Lee	xavier19@adventure-works.com	Road-150 Red, 48	1	3578.
11	SO44035	1	2019-07-29	Derrick Serrano	derrick15@adventure-works.com	Road-150 Red, 48	1	3578.
12	SO43747	1	2019-07-07	Christian Jackson	christian47@adventure-works.com	Road-150 Red, 56	1	3578.
13	SO43824	1	2019-07-15	Marcel Truempy	marcel0@adventure-works.com	Road-650 Black, 44	1	699.09
14	SO43985	1	2019-07-24	Cara Liang	cara12@adventure-works.com	Road-150 Red, 52	1	3578.
15	SO44006	1	2019-07-26	Lacey Guo	lacey30@adventure-works.com	Road-150 Red, 62	1	3578.

10,000+ rows | Truncated data | 2.64s runtime Refreshed 4 days ago

Aggregations

4 days ago (1s) 3

```
from pyspark.sql.functions import sum

sales_by_month = df.groupBy("order_year", "order_month") \
    .agg(sum("sales_amount").alias("monthly_sales")) \
    .orderBy("order_year", "order_month")

display(sales_by_month)
```

(2) Spark Jobs

sales_by_month: pyspark.sql.connect.dataframe.DataFrame = [order_year: integer, order_m

Table +

	order_year	order_month	monthly_sales
1	2019	7	927253.764200001
2	2019	8	515915.8711999998
3	2019	9	509929.4839999996
4	2019	10	561681.4757999999
5	2019	11	734261.5514000002
6	2019	12	596746.5568000001
7	2020	1	610280.5667999999
8	2020	2	580393.9611999998
9	2020	3	700873.1832000001
10	2020	4	632797.0313999997
11	2020	5	691261.3806
12	2020	6	485867.42400000006
13	2020	7	543819.9083000001
14	2020	8	363610.4897
15	2020	9	402246.7347999999

30 rows | 1.31s runtime

✓ 4 days ago (1s)

```python
from pyspark.sql.functions import sum, round

sales_by_year = df.groupBy("year") \
    .agg(round(sum("sales_amount"), 2).alias("total_sales"))

display(sales_by_year)
```

(2) Spark Jobs

sales_by_year: pyspark.sql.connect.dataframe.DataFrame = [year: integer, total_sales: d

Table +

	year	total_sales
1	2019	3845788.7
2	2020	6360375.84
3	2021	10686615.69

✓ 03:17 PM (1s) 5

```python
from pyspark.sql.functions import sum, round, desc

top_products = df.groupBy("product") \
    .agg(round(sum("sales_amount"), 2).alias("total_sales")) \
    .orderBy(desc("total_sales")) \
    .limit(10)

display(top_products)
```

(2) Spark Jobs

top_products: pyspark.sql.connect.dataframe.DataFrame = [product: string, total_sales: double]

Table +

	product	total_sales
1	Road-150 Red, 48	1202298.72
2	Road-150 Red, 62	1198720.45
3	Road-150 Red, 52	1077059.27
4	Road-150 Red, 56	1055589.65
5	Road-150 Red, 44	1001915.6
6	Mountain-200 Black, 46	925946.5
7	Mountain-200 Black, 42	842425.26
8	Mountain-200 Silver, 46	816222.19
9	Mountain-200 Silver, 38	815145.05
10	Mountain-200 Black, 38	799148.3

10 rows | 1.38s runtime

03:18 PM (2s) 6

```python
from pyspark.sql.functions import countDistinct

customer_growth = df.groupBy("year") \
    .agg(countDistinct("customer_name").alias("unique_customers"))

display(customer_growth)
```

(3) Spark Jobs

customer_growth: pyspark.sql.connect.dataframe.DataFrame = [year: integer, unique_custom

Table +

	year	unique_customers
1	2019	1196
2	2020	2727
3	2021	10461

3 rows | 1.70s runtime

03:18 PM (1s)

```python
top_customers = df.groupBy("customer_name") \
    .agg(round(sum("sales_amount"), 2).alias("total_spent")) \
    .orderBy(desc("total_spent")) \
    .limit(10)

display(top_customers)
```

(2) Spark Jobs

top_customers: pyspark.sql.connect.dataframe.DataFrame = [customer_name: stri

Table

	customer_name	total_spent
1	Larry Vazquez	10899.62
2	Kaitlyn Henderson	10896.22
3	Nichole Nara	10876.32
4	Kate Anand	10872.06
5	Margaret He	10841.22
6	Lawrence Alonso	10836.9
7	Terrance Rodriguez	10829.22
8	Rosa Hu	10817.6
9	Aaron Wright	10813.63
10	Clarence Gao	10799.52

10 rows | 1.43s runtime

Saved everything as delta tables in gold volume

5 minutes ago (7s)

```
sales_by_year.write.format("delta").mode("overwrite") \
    .saveAsTable("rithikadb.gold_schema.sales_by_year")
```

(2) Spark Jobs

4 minutes ago (3s)

```
sales_by_month.write.format("delta").mode("overwrite") \
    .saveAsTable("rithikadb.gold_schema.sales_by_month")
```

(4) Spark Jobs

4 minutes ago (3s)

```
top_products.write.format("delta").mode("overwrite") \
    .saveAsTable("rithikadb.gold_schema.top_products")
```

(2) Spark Jobs

4 minutes ago (2s)

```
top_customers.write.format("delta").mode("overwrite") \
    .saveAsTable("rithikadb.gold_schema.top_customers")
```

(2) Spark Jobs

4 minutes ago (3s)

```
customer_growth.write.format("delta").mode("overwrite") \
    .saveAsTable("rithikadb.gold_schema.customer_growth")
```

(3) Spark Jobs

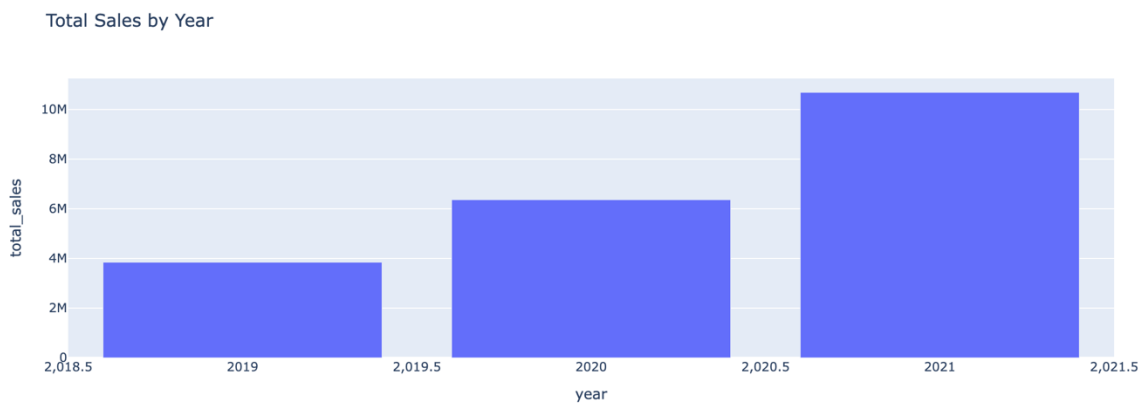
Visualization

4 minutes ago (4s) 14 Python

```python
import plotly.express as px

pdf = sales_by_year.toPandas()

fig = px.bar(pdf, x="year", y="total_sales",
             title="Total Sales by Year")

fig.show()
```

(2) Spark Jobs

pdf: pandas.core.frame.DataFrame = [year: int32, total_sales: float64]

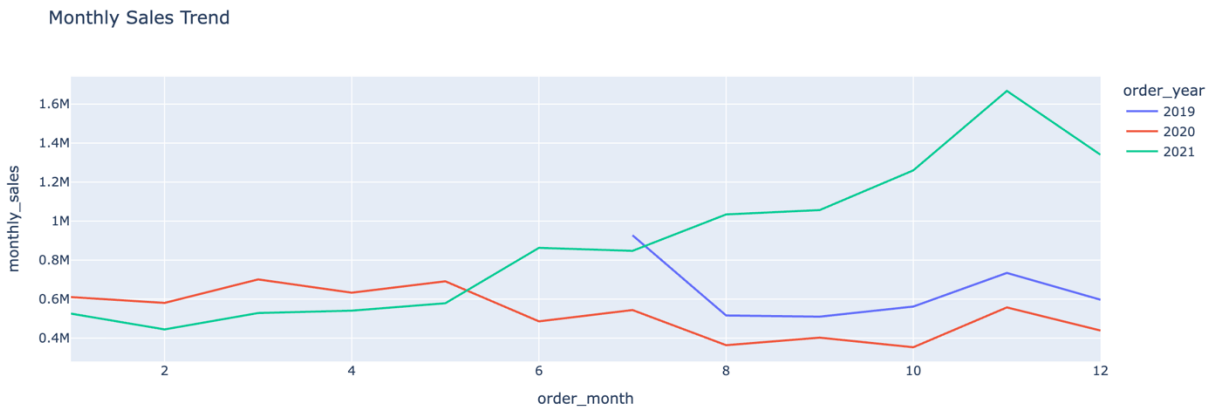


03:22 PM (1s) 15 Python

```python
pdf_month = sales_by_month.toPandas()

fig = px.line(pdf_month,
              x="order_month",
              y="monthly_sales",
              color="order_year",
              title="Monthly Sales Trend")

fig.show()
```

(4) Spark Jobs

pdf_month: pandas.core.frame.DataFrame = [order_year: int32, order_month: int32 ... 1 more field]

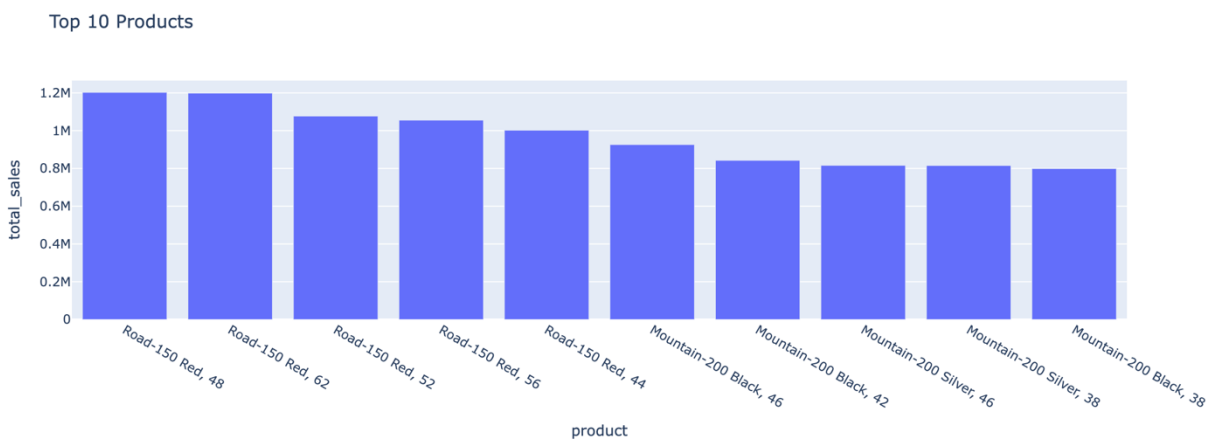


5 minutes ago (1s) 16

```
pdf_products = top_products.toPandas()

fig = px.bar(pdf_products,
             x="product",
             y="total_sales",
             title="Top 10 Products")

fig.show()
```

(2) Spark Jobs

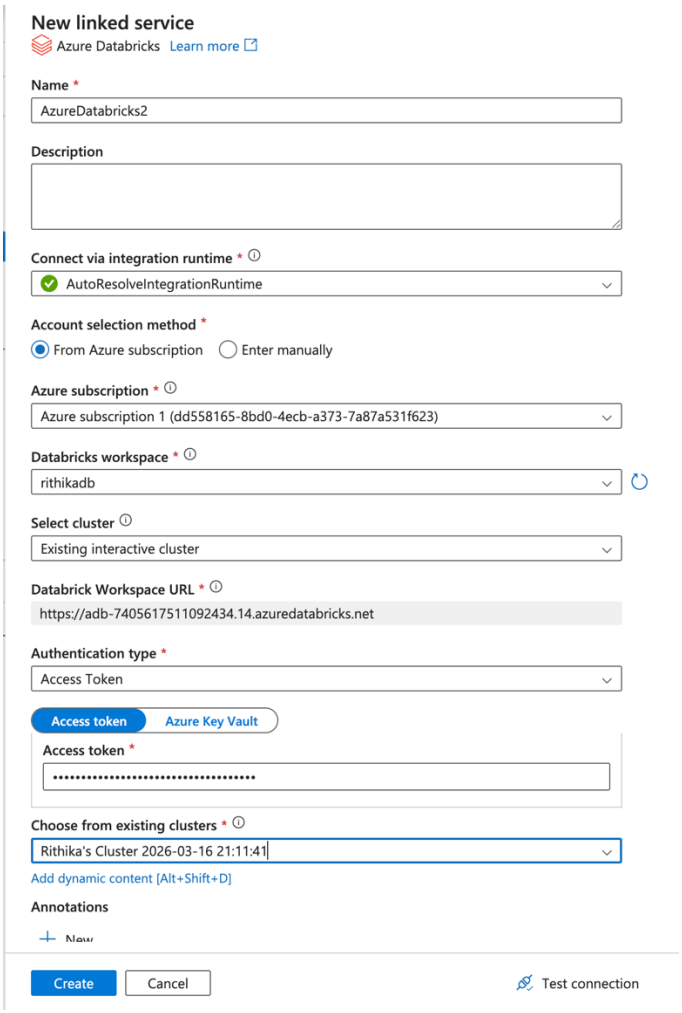
pdf_products: pandas.core.frame.DataFrame = [product: object, total_sales: float64]



ADF

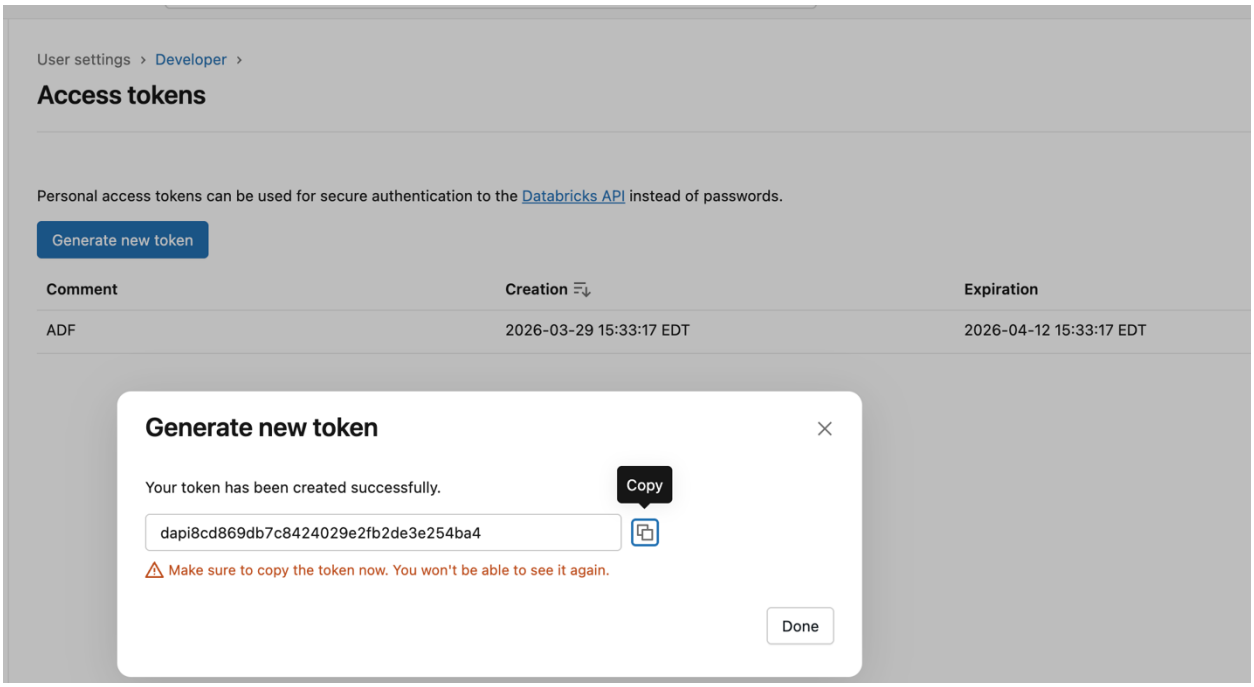
In data factory create a new pipeline and add three data brick notebooks

In the databricks activity under azure databricks tab create a new linked service.

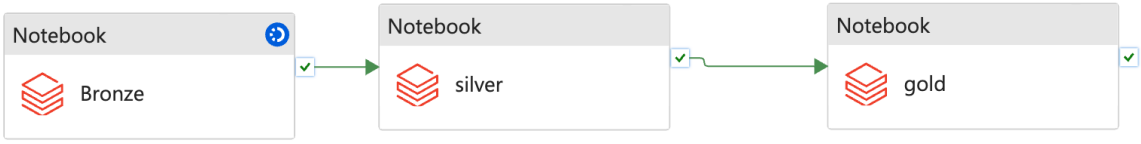


For the access token

- Go to databricks
- Click your profile
- Settings
- Developer tab
- Click access token
- Generate token
- Name your token and how many days the token should be used



Token: dapi8cd869db7c8424029e2fb2de3e254ba4

Then under settings tab select the databricks notebook you want to run.

Then connect all the 3 databricks activity and run it.